Section 4.5 Exercises

Directions: For exercises 1 - 6, use properties of logarithms to expand each of the following. Simplify as much as possible.

1. $\log_b(7x \cdot 2y)$

2. $\ln(3ab \cdot 5c)$

3. $\log_b\left(\frac{13}{17}\right)$

4. $\log_4\left(\frac{\frac{x}{z}}{w}\right)$

5. $\ln\left(\frac{1}{4^k}\right)$

6. $\log_2(y^x)$

Directions: For exercises 7 – 12, write each of the following as a single, simplified logarithm.

7. $\ln(7) + \ln(x) + \ln(y)$

8. $\log_3(2) + \log_3(a) + \log_3(11) + \log_3(b)$

9. $\log_b(28) - \log_b(7)$

10. $\ln(a) - \ln(d) - \ln(c)$

11. $-\log_b\left(\frac{1}{7}\right)$

12. $\frac{1}{3}\ln(8)$

Directions: For exercises 13 - 17, use properties of logarithms to expand each of the following. Simplify as much as possible.

13. $\log\left(\frac{x^{15}y^{13}}{z^{19}}\right)$

14. $\ln\left(\frac{a^{-2}}{b^{-4}c^5}\right)$

15. $\log\left(\sqrt{x^3y^{-4}}\right)$

16. $\ln\left(y\sqrt{\frac{y}{1-y}}\right)$

17. $\log\left(x^2y^3\sqrt[3]{x^2y^5}\right)$

Directions: For exercises 18 - 22, write each of the following as a single, simplified logarithm.

18. $\log(2x^4) + \log(3x^5)$

19. $\ln(6x^9) - \ln(3x^2)$

20. $2\log(x) + 3\log(x + 1)$

21. $\log(x) - \frac{1}{2}\log(y) + 3\log(z)$

22. $4\log_7(c) + \frac{\log_7(a)}{3} + \frac{\log_7(b)}{3}$

Directions: For exercises 23 - 24, rewrite each expression as an equivalent ratio of logs using the indicated base.

23. $\log_7(15)$ to base e

24. $\log_{14}(55.875)$ to base 10

Directions: For exercises 25 - 27, use properties of logarithms to evaluate without using a calculator.

25. $\log_3\left(\frac{1}{9}\right) - 3\log_3(3)$

26. $6\log_8(2) + \frac{\log_8(64)}{3\log_8(4)}$

27. $2\log_9(3) - 4\log_9(3) + \log_9\left(\frac{1}{729}\right)$

Directions: For exercises 28 - 32, use the change-of-base formula to evaluate each expression. Use a calculator to approximate each to three decimal places.

28. $\log_3(22)$

29. $\log_8(65)$

30. $\log_6(5.38)$

31. $\log_4\left(\frac{15}{2}\right)$

32. $\log_{\frac{1}{2}}(4.7)$